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Crowd Density Estimation via Markov Random Field (MRF)

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Abstract— **Crowd density estimation is of importance in security monitoring. Many crowd disasters happened because of the loss of control of the crowd density. This paper presents an algorithm to estimate crowd density by employing Markov Random Field (MRF). Three types of image features are extracted for estimating, and they are affected more by the neighboring features than by others, meeting the properties of Markov. The method of least squares is applied to estimate the model of crowd density. The system is applied for real-time videos. The proposed algorithm can estimate the number of people in crowds, and the experiments have shown the effectiveness.**

I. Introduction

With the improvement of living standards, people gather in public more and more, which results in many crowd accidents happening. To keep crowds in one place under management is an urgent task. For reducing the numbers of crowd accidents, estimating the crowd density should be very pivotal. As the crowd density is one of the essential description of all the status of crowd, it's convenient for us to learn the distribution of crowd and find out the tendency of abnormal behaviors of crowd, if we know more about the density of crowd.

There were much work dealing with estimating the density of crowd. Hartono Septian et al in [1] presented a novel approach to count the number of people that pass the view of an overhead mounted camera. B. Yang et al in [2] introduced a geometric algorithm that calculates bounds on the number of persons in each region of the projection, after phantom regions have been eliminated, and the approach is not based on the explicit detection of individuals in each image frame. Enwei Zhang et al in [3] employed group tracking to compensate weakness of multiple human segmentation, which can handle complete occlusion. Some surveillance systems [4], [5], [6] are good for classifying the crowd image into different levels of crowd density . Velastin et al in [7] proposed a structural method using edge or corner detection. Some image processing techniques applicable to the measurement of density and motion in crowded scenes have been presented. Ma et al in [8] proposed a pixel counting method based on foreground segmentation. And it investigated the geometric correction to account for perspective distortion. M.H. Pi et al in [9] proposed a two-stage image segmentation method, and it applied different image segmentation techniques. The methods proposed in [10] estimated the crowd density through calculating the number and position of the people in the image. Obviously head is one of the most important features for one person in the crowd. There are several head detection algorithm and most of them are based on appearance [10], knowledge [11], feature invariant [12], and template matching [13], [14]. The goal of this work [15] is to use computer vision to measure crowd density in outdoor scenes. In [16], the Haar wavelet transform was applied to extract the featured area of the head-like contour, and then the support vector machine was proposed to classify these features as a head or not. The perspective transforming technique of computer vision is used to estimate crowd size more accurately. And the system goes one step further to estimate the number of people in crowded scenes in a complex background by using a single image. Shengsheng Yu et al in [17] estimated the number of people passing a gate or a door provides useful information for video-based surveillance and monitoring applications. Two methods based on different approaches of texture analysis, one statistical and another spectral, were applied in [18].

In this paper, we present an algorithm to estimate crowd density by employing Markov Random Field (MRF) [19], [20]. MRF is a probability theory that promises to be important both in the theory and application of probability. It includes two level of meaning. Firstly, it has the Markov property, meaning that future states depend only on the present state, and are independent of past states. Secondly, random field is a generalization of a stochastic process such that the underlying parameter need no longer be a simple real, but can instead be a multidimensional vector space or even a manifold. Three types of image features including optical flow, foreground and edge were extracted for estimating the crowd density. The system is applied for real-world videos. Frow the point of view of position of people in the crowd, the image of crowd meet the properties of Markov. It means that the features are affected more by the neighboring features than by others. This is the reason for us to employ MRF in this paper. We can estimate the number of people in crowds by employing the proposed method, and the experiments show the effectiveness of the proposed system.

The rest of this paper is organized as follows. In section II, we describe the outline of our system. In section III, the process of extracting the features is explained. In section IV, we discuss the algorithm of estimating the crowd density applying MRF. In section V, the experimental results are shown and analyzed. In the last section, we summarize the approach and present some clues for future research work.

II. Overview of our crowd modeling system

The system includes two primary contents: 1) Pre-processing: optical flow computation, foreground detection, and edge detection. Three types of features were extracted in the first step. 2) Crowd density estimation employing Markov Random Field

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(MRF). And the reason for us choosing MRF is that the features we extracted meet the properties of Markov. Fig. 1 shows the overview of our crowd density estimation system.

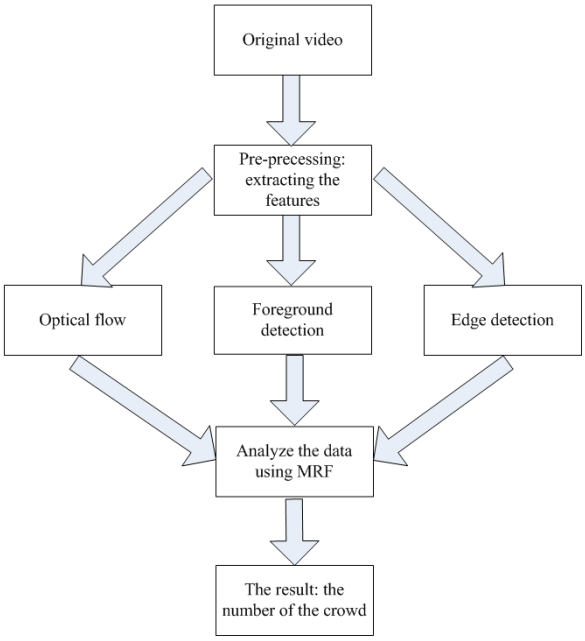


Fig. 1. Overview of our crowd density estimation system.

III. Pre-Processing: Extract the features

We extract three types of features to estimate the crowd density. The features include optical flow, foreground and edge.

A. *Optical flow*

As the pattern of apparent motion of objects, surfaces, and edges in a visual scene, optical flow is produced by the relative motion between an observer and the scene. Lucas-kanade algorithm is applied to calculate optical flow in this paper. However, optical flow is quite sensitive to noise. The traditionary optical flow is showed in Fig. 2. We can find in Fig. 2 that there is much noise in the result. We use the detected foreground to reduce the noise. This will be approached later.



Fig. 2. Optical flow of the frame

B. *Foreground detection*

In the matter of the system about video content analysis, foreground detection plays a very pivotal role. It is a foundation for object tracking, recognition, counting, and so on. Generally, there are two types of foreground detection methods. One is adaptive, the other is non-adaptive. Adaptive methods usually keep a background model fixedness and the parameters of the background model evolve with time. Non-adaptive methods depend on certain numbers of video frames and do not need a background model fixedness in the algorithm. In this paper, we use statistical foreground detection algorithm based on Gaussian model to reduce the noise pollution in the obtained optical flow as you can see in Fig. 3.

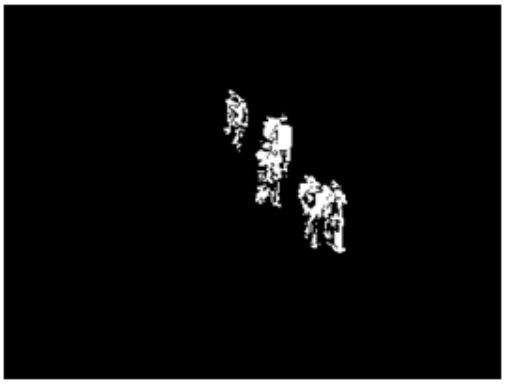


Fig. 3. Foreground of the frame

The noise is reduced as Fig. 4. We consider optical flow vectors only inside foreground areas by combining optical flow information with the foreground mask.



Fig. 4. Optical flow & Foreground detection

C. *Edge detection*

Edge detection is a terminology in image processing and computer vision, particularly in the areas of feature detection and feature extraction, to refer to algorithms which aim at identifying points in a digital image at which the image brightness changes sharply or more formally has discontinuities. Canny edge detection is used here as Fig. 5.



Fig. 5. Edge of the frame

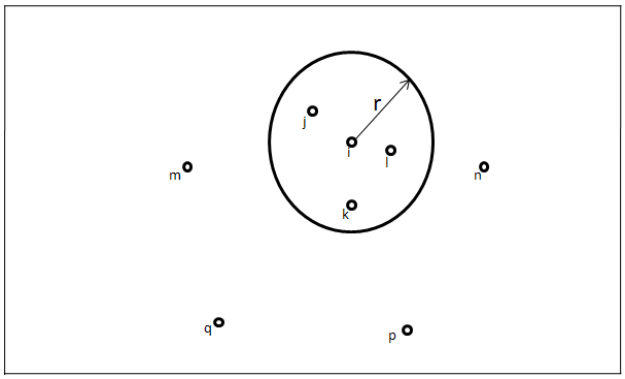


Fig. 6. An example of neighborhood system

IV. Crowd Density Estimation using Markov Random Field (MRF)

A. Markov Random Field

Markov random field [19],[20] is very important both in the theory and application of probability. It contains two level of meaning, the Markov property and the random field. The former means that future states depend only on the present state and independent of past states. The latter one means a generalization of a stochastic process. The prototypical model of MRF is the lsing model [19]. Random field theory [20] is good for dealing with many of the problems that we encounter in functional imaging. In probability theory, let states space $S = \{X_1, ..., X_n\}$, with the statesX_i in $\{0, 1, ..., K-1\}^n$ being a set of random variables on the sample space $\Omega = \{0, 1, ..., K-1\}^n$. The probability P is defined as a random field if for all ω in Ω,

$$P(\omega) > 0. \tag{1}$$

Given an graph G = (V, E), a set of random variables $\{X_v\}_{v \in V}$ indexed by V form a Markov random field if they satisfy the Markov property and the random field property.

$$P(X_i | X_{S-\{i\}}) = P(X_i | X_{N_i}) \tag{2}$$

Where N_i is a set of neighbors of the random variable X_i.

$$N_i = \{i^{'} \in S | [dist(pixel_i, pixel_{i^{'}})]^2 \leq r, i^{'} \neq i\} \tag{3}$$

In other words, the probability that a random variable assumes a value depends on the other random variables only through the ones that are its immediate neighbors.

For example, the neighbors of site i in Fig. 6 is {j, k, l}.

B. Crowd Density Estimation employing Markov Random Field

The abnormal behaviors of crowds are always relative with the crowd density. A number of surveillance applications require the estimation of crowd density to ensure security, safety, and site management. As crowd density increases, the overlap among crowd members gets worse. Moreover, there are significant varieties in color and texture of the crowd, and the backgrounds against which the people lie are unconstrained and complex. In this paper, MRF is employed here to estimate the density of the crowd. From the aspect of position of people in the crowd, the features meet the properties of Markov. And this is the right reason why we employ MRF in this paper.

The details of the processing of the crowd density estimation system based on MRF are as follows.

Input: Three kinds of features of the image

Output: Crowd density

i) Feature normalization based on the position.

ii)Smoothing the features.

iii) Based on the definition and properties of MRF, different neighbor values are used here to calculate the weights of each point of the features.

iv) Estimate the density of crowd in the image.

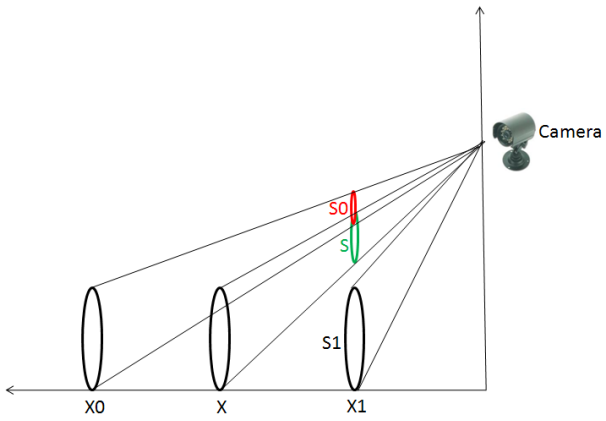


Fig. 7. Geometry of the objects

Because the objects will be smaller as their distance from the camera increases, we need do the normalization to the features. As Fig. 7 shown, we can analyze it as follows.

We choose two regions X0, X1 as we have known. We define the distance between them as d. The relation of them will be referenced as equation(10).

$$d = (X0 - X1) \tag{4}$$

The areas of them were defined as S0, S1. We can easily gain the proportion of them.

$$\frac{S0}{S1} = (\frac{X1}{X1+d})^2 \tag{5}$$

We take X1 as the standard region.

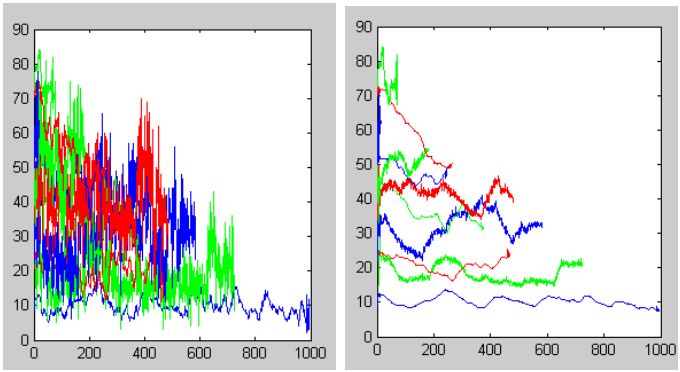
$$X1 = \frac{d}{\sqrt{\frac{S1}{S0}} - 1} \tag{6}$$

Define S as the area of region X. We can get the formula as

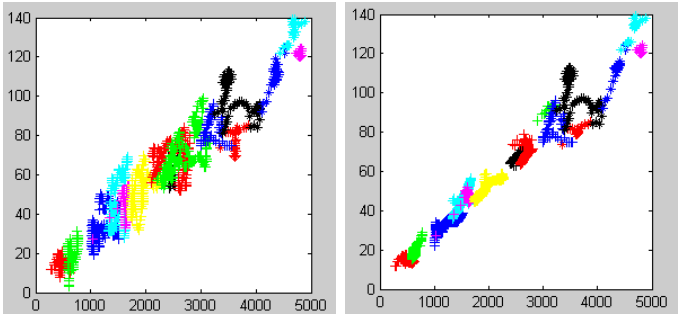
$$\frac{S}{S1} = (\frac{X1}{X})^2 \tag{7}$$

Finally,we obtain the function of normalization f(X) as follows.

$$f(X) = (\frac{X1}{X})^2 = (\frac{d}{(\sqrt{\frac{S1}{S0}} - 1) * X})^2 \tag{8}$$



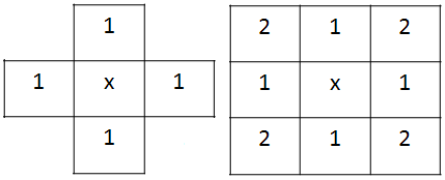
(a) Before smoothing (b) After smothing



(c) Before smoothing (d) After smothing

Fig. 8. Smoothing

For reducing the influence of noise of the features, we need take a smoothing method before we do the next step. In this paper, we used mean value to smooth the noise. The smoothing process is as Fig. 8. In Fig. 8.(a), x-coordinate indicates the sample number, and y-coordinate indicates the features value of image. And the points of different color indicate different number of people. From the bottom to top, the number increases from one to ten. Fig. 8.(b) shows the smoothing result of Fig. 8.(a). In Fig. 8.(C), x-coordinate and y-coordinate indicate the different kinds of features value of image. And the points of different color indicate different number of people. Fig. 8.(d) shows the smoothing result of Fig. 8.(c).



(a) A set of neighbours (b) A set of neighbours

Fig. 9. Neighbours

6	4	3	4	6
4	2	1	2	4
3	1	x	1	3
4	2	1	2	4
6	4	3	4	6

Fig. 10. A set of neighbours

Then different neighbor values are proposed here to calculate the weights of each point of the features. In this paper, we applied the three neighbor values shown in Fig. 9 and Fig. 10.

The method of least squares is applied in this paper to estimate the model of crowd density. Let $f_i\{i = 1,2,...N\}$ be the number of features of the ith frame, t_i be the true number of people in the frame, whose value is counted by us. And let function $ER(f,\alpha)$ be the experiment result for the features. The vector α represents the parameters of function ER. The least squares method finds the optimum model of crowd density when the sum, S, of squared residuals is a minimum.

$$S = \sum_{i=1}^{n} r_i^2 \tag{9}$$

Where r_i represents the residual. A residual is defined as the difference between the true number of people and the experiment result of number of people in the frame.

$$r_i = t_i - ER(f_i,\alpha) \tag{10}$$

The method of least squares is simple and effective for our experiments, so it's applied in our system. The experiments results of it are shown in the next section.

C. Crowd groups definition

Defining crowd groups is a very pivotal assignment. Before counting the number of people in crowds, we should define what is crowd groups. We can estimate the crowd density more accurately with better crowd groups definition.

J.N. Guo et al in [21] proposed an algorithm to defining crowd groups. It defined crowd groups by analyzing crowd flow direction and their position. The result of one example is showed in Fig. 11. The four people walk together, and we define them as a crowd group.



Fig. 11. Crowd groups definition.

The primary steps are as follows.

1) Based on crowd flow direction and their position which we already know [21], we define original crowd with the people who are close to crowd flow direction in vertical scope.

2) Filter the people who are inconsistent with the crowd flow direction.

3) Filter the people in crowd flow direction who are far away from the great mass of other people in the crowd.

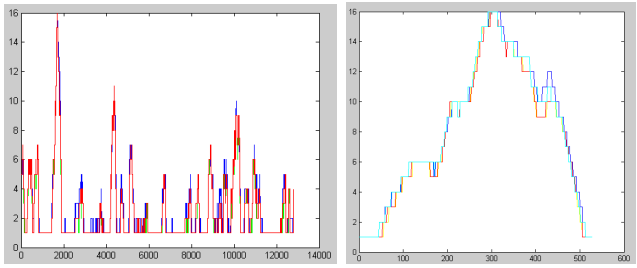
The detailed description is shown in [21] and the experiments result is shown in next section.

V. Experiment Result

In our experiments, we aim to estimate the crowd density. We run our system on the same scenes in our campus. To estimate the crowd density, we need to obtain the features beforehand. Three features we captured are optical flow, foreground, and edge. Moreover, we use the detected foreground to filter the noise of the optical flow. As the objects will be smaller as their distance from the camera increases, we do the feature normalization based on the position. Then we use different neighbor values to calculate the weights of each point of the features, which is based on Markov Random Field. All the videos used in our experiments are 320*240 pixels.

Combining the crowd density and crowd group definition, our results are shown in Fig. 12, Fig. 13, Table I. There were much work dealing with estimating the density of crowd, but few of them combined the two factors. The analysis of the experiments is as follows.

In Fig. 12 (a), it shows results of all frames. We take some frames of them to analyze as Fig. 12 (b). There are four different lines in Fig. 12 (b), the cyan line represents the true number of people. And the other three line, including blue line, yellow line, and red line, represent the experiment results with different neighbors values which are shown in Fig. 9 and Fig. 10. For more details, we analyze the exact frame as Fig. 13. What the Fig. 13 (a) shown is that three persons walk together with one person not in their group. What the Fig. 13 (b) shown is that five persons go in a group. In Fig. 13 (c) and Fig. 13 (d), because the number of people is relatively more than the former, the result is sometimes not so accurate as what you see. There are three numbers in the image. The first one is the result based on the neighbors value in Fig. 9 (a), the second one is based on the neighbors value in Fig. 9 (b), and the last one is based on the neighbors value in Fig. 10. The results of our experiments demonstrate that our algorithm can estimate the crowd density and perform crowd group definition well.

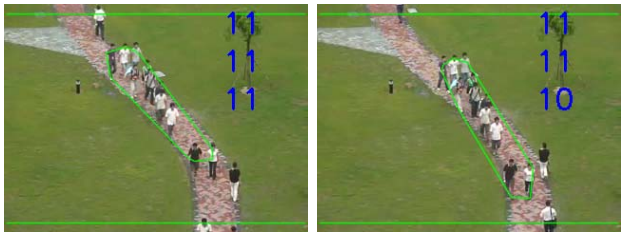


(a) All frames (b) Some frames

Fig. 12. Experiment results



(a) (b)



(c) (d)

Fig. 13. Results of each frame

In Table I, it shows the accuracy rate of our experiments. It's about the relation between true number of people in the crowd and the experimental result. In this table, M represents the true number of people. N represents the frames of special M. ER represents the experimental result. In most cases, the accuracy rate is decent. And even in bad cases, the experimental results are not too bad. The larger of the values of N, the more acceptable of the experiment result, as N stands for the sample number.

According to and Fig. 12, Fig. 13, Table I, we conclude that the algorithm we employed in this paper can effectively count the number of people in the crowd. But then, as the number of people is not very large, we can not make sure of that the

TABLE I

ACCURACY RATE OF OUR EXPERIMENTS. M REPRESENTS THE TRUE NUMBER OF PEOPLE. N REPRESENTS THE FRAMES OF SPECIAL M. ER REPRESENTS THE EXPERIMENTAL RESULT.

M(N)	1(3447)	2(1904)	3(1200)	4(957)	5(1108)	6(814)	7(417)	8(479)	9(483)	10(267)	11(45)	12(72)	13(78)
ER=1	96.4%	4.3%											
ER=2	3.6%	80.1%	7.3%										
ER=3		15.6%	81.6%	6.8%									
ER=4			11.1%	86.9%	5.6%								
ER=5				6.3%	89.0%	8.0%							
ER=6					5.4%	86.5%	8.9%						
ER=7						5.5%	88.5%	5.0%					
ER=8							2.6%	90.0%	7.9%				
ER=9								5.0%	84.4%	3.4%			
ER=10									7.7%	86.9%	4.5%		
ER=11										9.7%	73.3%	1.4%	
ER=12											22.2%	72.2%	5.1%
ER=13												26.4%	82.1%
ER=14													12.8%

proposed algorithm can keep its effectiveness in the case of crowded people.

VI. CONCLUSIONS AND FUTURE WORK

In this paper, we propose an intelligent surveillance system based on Markov Random Field to estimate the density of crowd. We extract three kinds of features including optical flow, foreground and edge. The features are effected by the neighboring features much more than others, which meets the properties of markov. It is the reason for us employing MRF in this paper. Real-world videos are applied for our experiments. The proposed system can estimate the number of people in crowd effectively in the case of relatively small number of objects. In future, based on what we have done, we will focus on crowd abnormity detection.

VII. ACKNOWLEDGMENTS

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